FamilyForge — User Manual

A pedigree-aware clinical variant-interpretation tool (annotated VCF + pedigree + panel → ranked candidates + clinician / publication / clinical reports)

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1. Introduction

FamilyForge is a pure-Python, disease-agnostic **clinical variant-interpretation tool**. It takes a single family's annotated multi-sample VCF, a pedigree, and a curated gene panel, and produces **segregation-checked candidate variants**, a pragmatic **ACMG-style A/B/C tiering**, a **copy-number (CNV) cross-check**, genotype-based **relatedness / sex QC**, and a set of **audience-specific reports** (a concise clinician report, a full publication report, and a letterhead clinical-genomics report).

FamilyForge is the **interpretation** member of the *forge* family of sibling pipelines:

> **CaptureForge** (probe *design*) → **CallForge** (sequence *analysis*) → **FamilyForge** (family *interpretation*).

CallForge turns demultiplexed FASTQs into a joint-genotyped, annotated callset; FamilyForge consumes that callset — specifically the annotated joint `VCF.gz`, and optionally CallForge's `cnv_calls.tsv` and `provenance.json` — and answers the clinical question CallForge does not: *given this family and this panel, which variants best explain the affected members' disease, and how strong is the evidence?*

What makes it "family-aware." FamilyForge never scores a variant in isolation. Every candidate is tested against the **affected / unaffected structure of the pedigree** under explicit Mendelian models, cross-checked for **monozygotic-twin concordance**, and — where a parent was not sequenced — labelled honestly as *inferred, not confirmed*. Population frequency thresholds are applied differently for dominant versus recessive hypotheses, and the tiering rubric folds in ClinVar, predicted consequence, in-silico damage, conservation, and the gene's known disease mechanism.

Who it is for. Clinical bioinformaticians, medical geneticists, and genomics-core staff who need a reproducible, honest first pass over a small family's exome/genome data — one that ranks candidates, shows its evidence trail, and produces documents suitable for a multidisciplinary review or a case report.

> **Scope and status.** FamilyForge is a **research** tool. Its outputs are *candidates*, not diagnoses: every Tier A/B finding is explicitly pending orthogonal confirmation (Sanger / MLPA / aCGH), functional validation, and formal ACMG classification by an accredited laboratory before any clinical action. It is **implemented and verified end-to-end on a synthetic self-test** (§12). The bundled gene-specific rules and the default report framing were developed around a hyper-IgE-syndrome (HIES) inborn-error-of-immunity use case, but the engine is disease-agnostic: the **panel and pedigree are inputs**, never hardcoded. §11 explains exactly which parts are generic and which carry the HIES framing.

2. Concepts

This section defines the ideas the tool is built on. It is worth reading once before the command reference, because the flags map directly onto these concepts.

2.1 The pedigree / phenotype model

FamilyForge represents a family as a set of **members**, each parsed from one line of a standard **6-column PED** file:

Column	Meaning	Notes
1	Family ID	free text (e.g. `FAM01`)
2	Individual ID (`iid`)	**must match the VCF sample name** for a sequenced member
3	Father ID	`0` if unknown/founder
4	Mother ID	`0` if unknown/founder
5	Sex	`1`/`M`=male, `2`/`F`=female, `0`/`U`=unknown
6	Phenotype	`2` = affected; anything else = unaffected

A crucial design choice: a member is marked **sequenced** *if and only if* its `iid` appears among the VCF sample names. Members present only to define relationships — most commonly an **unsequenced father** — are kept in the pedigree so the models can reason about obligate carriers and phase, but they contribute no genotypes. This is what lets FamilyForge state, for a de-novo-pattern variant, "*de novo OR paternally inherited (unconfirmed)*" rather than over-claiming.

Derived quantities the models use: `affected` (sequenced affected members), `unaffected` (sequenced unaffected members), and `mothers_of_affected()` (the set of mothers, present in the pedigree, of the affected members — the obligate-carrier check for recessive models).

2.2 Genotype quality gating

Before any inheritance model runs, each per-sample genotype is gated. A het, hom-alt, or hemizygous-alt call is demoted to "not usable" (`ok = False`, treated like a no-call by the models) when it fails the configured thresholds:

- genotype quality `GQ < min_gq` (default **20**), or
- depth `DP < min_dp` (default **8**), or
- for a heterozygous call, alt allele-balance `AB < min_alt_ab` (default **0.20**).

Homozygous-alt allele balance is handled leniently (a small residual REF fraction does not by itself void a hom-alt call). Zygosity itself is computed allele-count-wise, with **hemizygous** recognised on chrX/chrY for male samples. This gate is the reason a noisy low-depth genotype cannot, on its own, manufacture a de-novo or compound-het call.

2.3 Inheritance models

FamilyForge evaluates four single-variant / gene-level Mendelian models. Each is checked against the affected/unaffected structure; a model only fires if the panel's declared inheritance for the gene is compatible (or the panel leaves inheritance blank, in which case any model may fire).

- **De novo / dominant** (`de_novo_or_dominant`). All sequenced affected carry the allele (het), all sequenced unaffected are homozygous-reference, and the gene allows `AD` (or has no declared inheritance). If an unaffected relative also carries the allele, a separate *reduced-penetrance* dominant branch fires instead, at low confidence and flagged.
- **Autosomal-recessive homozygous** (`ar_homozygous`). All affected are homozygous-alt on an autosome, no unaffected sibling is homozygous, and (where sequenced) the mother is not homozygous-alt — an obligate-carrier mother is expected to be heterozygous. Confidence is *high* only when the twin genotypes are concordant and every sequenced mother is het.
- **Autosomal-recessive compound heterozygous** (`ar_compound_het`). Evaluated per gene: two different heterozygous variants, both het in *all* affected, on opposite parental haplotypes (*in trans*). Phase is inferred from the mother — one allele maternal (mother het), the other absent in the mother (mother hom-ref, therefore inferred paternal). No unaffected relative may carry **both** alleles. When the father is unsequenced, the paternal allele is labelled *inferred*.
- **X-linked recessive** (`x_linked_recessive`). On chrX, affected males are hemizygous for the allele, no sister is homozygous, the mother (where sequenced) is a heterozygous carrier, and the gene allows `XLR`/`XL`.

Each model emits a human-readable **segregation string** — e.g. `MOTHER(unaff)=het; SIBLING(unaff)=ref; TWIN_1(aff)=hom; TWIN_2(aff)=hom` — so a reviewer can see the family pattern at a glance.

2.4 Monozygotic-twin handling

If two or more samples are declared an **MZ-twin group** (via `--mz-twins` or config), they are treated as *genetic replicates*: for a candidate to reach high confidence, the twins must be **genotype-concordant** at that site. A discordant genotype across an MZ pair is almost always a technical artifact (a dropout or a miscall), so the candidate is flagged and its confidence capped — and in tiering, a low-confidence Tier A is demoted to Tier B. Independently, the genotype-QC step reports pairwise concordance across the whole panel, so you can *verify* that a declared MZ pair really behaves like one ($\approx$1.0) and that first-degree relatives sit near 0.5–0.7.

2.5 ACMG-style tiering (A / B / C)

FamilyForge assigns each candidate a **tier**, a pragmatic triage signal — *not* a full, formal ACMG classification:

- **Tier A — likely actionable.** ClinVar pathogenic/likely-pathogenic (rare, segregating); or biallelic predicted-LoF in a LoF-mechanism gene under a recessive model; or a STAT3 dominant-negative hotspot missense/LoF under a de-novo/dominant pattern; or a rare damaging missense in a Tier-1/2 gene with strong in-silico consensus (AlphaMissense likely-pathogenic, REVEL ≥ 0.7, or CADD ≥ 25) under a matching-inheritance model.

- **Tier B — strong VUS.** Predicted-LoF in a panel gene, or a rare missense with supportive in-silico evidence, where mechanism or inheritance still needs confirmation. Tier A findings are **capped to B** when ClinVar is *conflicting*, or when genotype confidence is low / MZ-twin discordant.
- **Tier C — weaker / downgraded.** Rare in-panel variants with limited support, or variants downgraded because they are too common in gnomAD for their model, or ClinVar benign.

The evaluation is top-to-bottom, first-match-wins; §9 and the tiering diagram (Figure 2) document every branch. Alongside the tier, FamilyForge records an **evidence trail** — the list of criteria that fired (ClinVar status, consequence, AF, SIFT/PolyPhen/PhyloP, AlphaMissense/REVEL/CADD, segregation model) — which is printed verbatim in the reports.

A separate `classify()` step maps each row to a five-level **colour convention** for the reports — **P** (pathogenic), **LP** (likely pathogenic), **VUS**, **LB** (likely benign), **B** (benign) — from ClinVar, tier, and allele frequency.

2.6 CNV interpretation

SNV/indel calling is blind to large deletions, yet deletions are a classic disease mechanism for some genes (the bundled example is **DOCK8** in AR-HIES). FamilyForge reads CallForge's `cnv_calls.tsv` (tolerant of column-name variations), keeps **deletions that overlap a panel gene**, and tiers them: **Tier A** if the gene is a configured CNV-priority gene *and* the event is in an affected sample, otherwise **Tier B**. Low-confidence CNV labels and the SNV-blind mechanism are surfaced as flags, and every CNV candidate carries the standing recommendation to confirm by read-depth/IGV and an orthogonal assay (MLPA/aCGH).

2.7 Genotype-based relatedness & sex QC

Independently of any external sites file (no somalier required), FamilyForge cross-checks the samples from their own genotypes:

- **Pairwise concordance** over shared, quality autosomal sites — a relatedness / MZ / sample-swap check.
- **Sex inference** from chrX heterozygosity — a sample with a very low chrX het-rate (< 0.10) is inferred male, otherwise female — compared against the declared sex in the PED.

These land in every report's QC section and in `candidates.json`.

3. Architecture

FamilyForge is a small, readable Python package. A single command (`familyforge`) orchestrates the whole run; a second entry point (`familyforge-clinical-report`) renders the letterhead clinical report from the machine-readable output. The figure below is the authoritative map: inputs (top) flow through ingest/parse, into the interpretation modules, into reporting, and out to the deliverables.

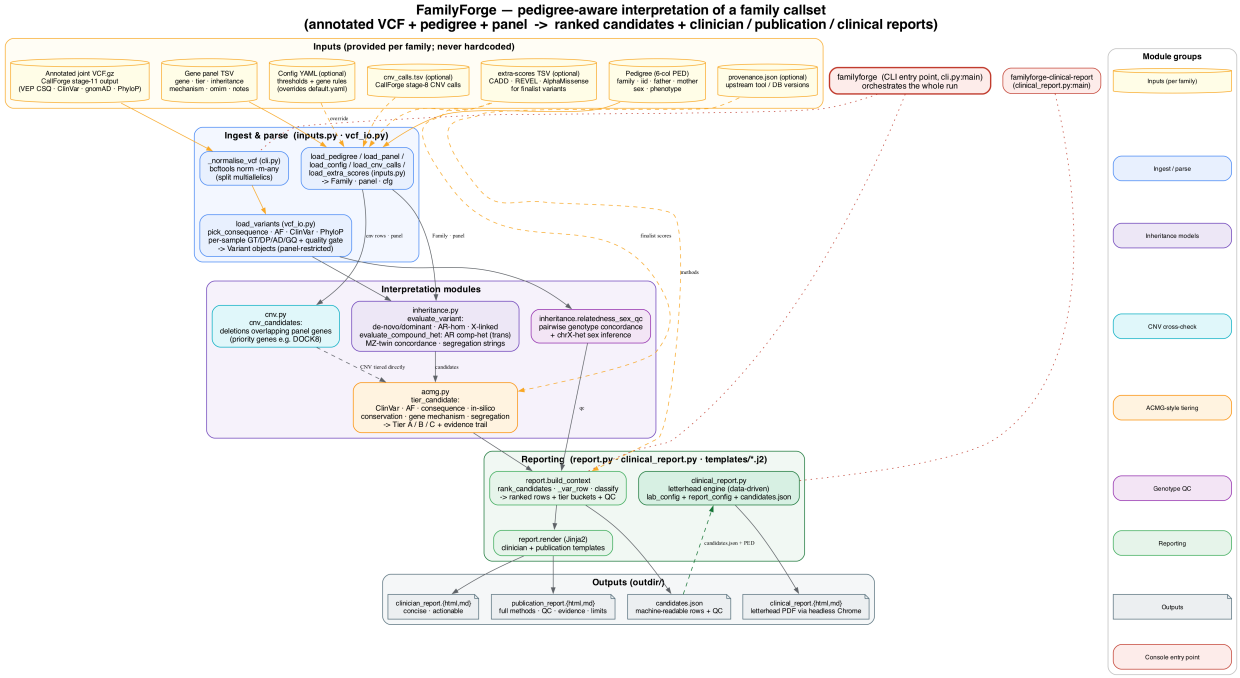


Figure 1: FamilyForge architecture — inputs (annotated VCF, pedigree, panel, optional config / CNV / extra-scores / provenance) are parsed by `vcf_io.py` and `inputs.py`, interpreted by `inheritance.py`, `cnv.py`, and `acmg.py`, then rendered by `report.py` and the `clinical_report.py` letterhead engine into clinician, publication, clinical, and JSON outputs.

Module map.

File	Responsibility
`cli.py`	Argument parsing, orchestration, VCF normalisation, JSON dump
`inputs.py`	Loaders: config (YAML), pedigree (PED), panel (TSV), CNV calls, extra scores
`vcf_io.py`	Parse the annotated VCF into `Variant` objects; pick consequence; quality-gate genotypes
`model.py`	Dataclasses: `Member`, `Family`, `GeneAnno`, `Variant`, `Candidate`
`inheritance.py`	Mendelian models, compound-het, MZ concordance, relatedness/sex QC

File	Responsibility
`acmg.py`	ACMG-style A/B/C tiering with an evidence trail
`cnv.py`	CNV deletion cross-check against panel genes
`report.py`	Build the report context; render clinician + publication (HTML + MD)
`clinical_report.py`	Standalone letterhead clinical-report engine (HTML + MD)
`config/default.yaml`	Default thresholds and gene-specific rules
`templates/*.j2`	Jinja2 templates for all four report styles
`tests/selftest.py`	Synthetic end-to-end test with planted, known-answer variants

Data-flow spine (as in `cli.main`): normalise VCF → load panel → discover which pedigree members are in the VCF → `load_variants` (panel-restricted, quality-gated) → apply optional `--extra-scores` → run `evaluate_variant` + `evaluate_compound_het` + `cnv_candidates` → `tier_candidate` each SNV/indel candidate → `relatedness_sex_qc` → `build_context` → `render` (four report files) → write `candidates.json`.

4. Installation

FamilyForge is a standard Python package (build backend: setuptools). It needs **Python ≥ 3.10** and five pure-Python/compiled dependencies: `cyvcf2`, `pandas`, `numpy`, `jinja2`, `pyyaml`. `cyvcf2` provides fast VCF parsing (it ships binary wheels for common platforms). **`bcftools` on `PATH`** is recommended but optional — it is used to split multiallelic sites; without it, multiallelic records are skipped with a warning.

4.1 pip + virtualenv (recommended for a single user)

```
# 1. obtain FamilyForge and enter the repository
git clone https://github.com/aduton1000/familyforge.git
cd familyforge

# 2. create and activate an isolated environment
python3 -m venv .venv
source .venv/bin/activate        # Windows: .venv\Scripts\activate

# 3. install (editable so local edits take effect immediately)
pip install -e .

# 4. confirm the console entry points are on PATH
familyforge --help
familyforge-clinical-report --help
```

4.2 conda / mamba (recommended when you also want bcftools)

```
mamba create -n familyforge -c conda-forge -c bioconda \
    python=3.11 cyvcf2 pandas numpy jinja2 pyyaml bcftools
mamba activate familyforge
pip install -e .          # register the familyforge console commands
```

4.3 Verify the installation

Run the bundled self-test (§12). It fabricates an annotated VCF with planted variants and confirms FamilyForge recovers the expected models, tiers, and reports:

```
python tests/selftest.py
# expect: Tier A genes: {'STAT3', 'DOCK8', 'PGM3'} ... SELFTEST PASS
```

5. Inputs

FamilyForge takes one required trio — **VCF + PED + panel** — plus four optional inputs. All are provided at run time; nothing is hardcoded.

5.1 The annotated VCF (`--vcf`, required)

A multi-sample, bgzipped VCF — normally CallForge's stage-11 `annotated.vcf.gz`. FamilyForge reads:

- **Per-sample FORMAT**: `GT`, `DP`, `AD`, `GQ` (for zygosity + quality gating).
- **VEP `CSQ`** (INFO): `Consequence`, `SYMBOL`, `HGVSc`, `HGVSp`, `SIFT`, `PolyPhen`, `CANONICAL`, `MANE_SELECT`/`MANE`, and the population-AF fields listed in `af_fields`.
- **INFO annotations**: `ClinVar_CLNSIG` (or `CLNSIG`, or the CSQ `CLIN_SIG` field), `PhyloP`, and `PARALOG_GENE`/`PARALOG_CONF` if present.

For each variant, FamilyForge **picks one transcript consequence** — preferring a panel gene, then a MANE/canonical transcript, then the most-severe consequence — and, in its primary pass, **keeps only variants whose picked gene is in the panel**. The MANE Select RefSeq accession is retained so the reports can present clinical-standard HGVS.

Multiallelic handling: FamilyForge runs `bcftools norm -m-any` first (unless `--skip-norm`), so genotype interpretation is unambiguous. If a record is still multiallelic at parse time, it is skipped.

5.2 The pedigree (`--ped`, required)

A 6-column PED (§2.1). Sample IDs in column 2 must match the VCF sample names for sequenced members. Example (`family.ped`) — an unsequenced father, an unaffected mother, an unaffected sibling, and affected MZ twins:

```
FAM01  FATHER   0       0       1  0
FAM01  MOTHER   0       0       2  1
FAM01  SIBLING  FATHER  MOTHER  2  1
FAM01  TWIN_1   FATHER  MOTHER  1  2
FAM01  TWIN_2   FATHER  MOTHER  1  2
```

Here `FATHER` has no genotypes in the VCF, so FamilyForge marks him *relationship-only*; `TWIN_1`/`TWIN_2` are the affected (phenotype `2`) males.

5.3 The gene panel (`--panel`, required)

A tab-separated file, one gene per row, with a header line beginning `gene`. Columns after the gene are optional; missing columns default sensibly.

```
gene    tier    inheritance    mechanism    omim      notes
STAT3   1       AD             DN           147060    DBD/SH2 hotspot incl R382
DOCK8   1       AR             LOF          243700    biallelic LOF, large
 ↪ deletions
```

```
PGM3    2       AR              HYPO          615816    hypomorphic/LOF
IL2RG   1       XLR             LOF           300400    X-linked SCID/IEI
```

- **tier** — the gene's a-priori priority (`1/2` = Tier-1/2 disease gene; used by the strong-missense Tier A branch). Defaults to `3`.
- **inheritance** — a comma-separated set of `AD`, `AR`, `XLR`/`XL`; gates which models may fire. Leave blank to allow any model.
- **mechanism** — free set (e.g. `LOF`, `DN`, `HYPO`); informational, surfaced in reports.
- **omim**, **notes** — free text for the reports.

5.4 Configuration (`--config`, optional)

A YAML file that **overrides** keys in the packaged `config/default.yaml` (§8). Supply only the keys you want to change.

5.5 CNV calls (`--cnv`, optional)

A CallForge `cnv_calls.tsv`. The parser is tolerant of column-name variants: it recognises `gene/genes/symbol/overlapping_genes`, `call/type/cnv/event/status`, `cn/copy_number/log2`, `sample/sample_id`, `chrom/chr`, `start`, `end`, `callability/confidence/label`. A row counts as a deletion if its call text says del/loss/deletion, or copy number `< 1.5`, or log2-ratio `< -0.4`.

5.6 Finalist in-silico scores (`--extra-scores`, optional)

When genome-wide dbNSFP is unavailable, you can attach externally retrieved scores for a handful of finalist variants. A TSV keyed by `chrom:pos:ref>alt`:

```
key                     cadd_phred    revel    alphamissense_score
 ↪  alphamissense_class
chr17:42322000:G>A      29.4          0.86     0.978
 ↪  likely_pathogenic
chr9:300000:C>T         34.0          .        .                       .
```

Matching rows have their CADD/REVEL/AlphaMissense fields populated before tiering, which can lift a strong missense to Tier A.

5.7 Provenance (`--provenance`, optional)

A CallForge `provenance.json`. Its contents (tool/DB versions, git commit) are threaded into the publication report's methods section for reproducibility.

6. The `familyforge` command

`familyforge` is the primary entry point. It runs the full interpretation in one call.

```
familyforge --vcf VCF --ped PED --panel PANEL --outdir DIR
            [--cnv TSV] [--config YAML] [--study-name NAME]
            [--mz-twins "ID1,ID2" ...] [--provenance JSON]
            [--extra-scores TSV] [--skip-norm]
```

Flag reference.

Flag	Req.	Purpose
`--vcf`	Yes	Annotated joint VCF.gz (CallForge stage-11 output)
`--ped`	Yes	6-column PED pedigree
`--panel`	Yes	Gene-panel TSV (`gene,tier,inheritance,mechanism,omim,n`
`--outdir`	Yes	Output directory (created if absent)
`--cnv`		CallForge `cnv_calls.tsv` for the CNV cross-check
`--config`		Override YAML (merged over `config/default.yaml`)
`--study-name`		Report title (default `Family study`)
`--mz-twins`		Comma-separated MZ-twin sample IDs; **repeatable** for multiple twin sets
`--provenance`		CallForge `provenance.json` for the methods section
`--extra-scores`		TSV of finalist CADD/REVEL/AlphaMissense scores
`--skip-norm`		Skip `bcftools norm` (use if the VCF is already biallelic-split)

Generic example.

```
familyforge \
  --vcf        /path/to/callforge/anno/annotated.vcf.gz \
  --ped        /path/to/config/family.ped \
  --panel      /path/to/config/gene_panel.tsv \
  --cnv        /path/to/callforge/cnv/cnv_calls.tsv \
  --config     /path/to/config/familyforge.yaml \
  --extra-scores /path/to/config/finalist_scores.tsv \
```

```
  --provenance /path/to/callforge/provenance.json \
  --mz-twins   TWIN_1,TWIN_2 \
  --study-name "Family FAM01 exome" \
  --outdir     /path/to/results/reports
```

What it prints (to stderr): the number of panel variants loaded and samples seen, how many extra scores matched, and the tier counts (`nA / nB / nC`), followed by the list of files written.

7. The `familyforge-clinical-report` command

The second entry point renders a **letterhead clinical-genomics report** — the disease- agnostic, data-driven document intended for a clinical file. It consumes the machine- readable `candidates.json` produced by a `familyforge` run, plus the pedigree and two small YAML files, and emits an HTML report (render to PDF with headless Chrome) and a Markdown twin.

```
familyforge-clinical-report --lab-config LAB.yaml --report-config REPORT.yaml
                            --candidates candidates.json --ped PED --out OUT.html
```

Flag	Purpose
`--lab-config`	Reusable lab branding YAML (institute, logos, head-of-lab, confidentiality)
`--report-config`	Per-report YAML (accession, specimen, disease name, sample labels, QC rows, management)
`--candidates`	The `candidates.json` from a `familyforge` run
`--ped`	The same PED (for sex/relationship fallback)
`--out`	Output HTML path (a `.md` twin is written alongside)

The `--lab-config` supplies branding fields (`lab_abbrev`, `institute`, `university`, `lab_name`, `logo_banner`, `logo_lab`, `head_name`, `head_title`, `confidentiality`); the logos are read and embedded as base64 so the HTML is self-contained. The `--report-config` supplies the per-case content: `accession`, `report_type`, `status`, `test`, `specimen`, `reference_genome`, `referral_reason`, `disease_name`, `panel_name`, a `samples` map (labels/relationships/status), optional `qc_rows`, `gene_management`, and `notes`. All clinical content is data — nothing about the disease is baked into the engine.

Generic example (render to PDF at the end):

```
familyforge-clinical-report \
  --lab-config    /path/to/config/lab_branding.yaml \
  --report-config /path/to/config/report_meta.yaml \
  --candidates    /path/to/results/reports/candidates.json \
  --ped           /path/to/config/family.ped \
  --out           /path/to/results/reports/clinical_report.html

# render the letterhead HTML to PDF (self-contained; repeating header per page)
chrome --headless=new --no-pdf-header-footer \
       --print-to-pdf=/path/to/results/reports/clinical_report.pdf \
       file:///path/to/results/reports/clinical_report.html
```

The letterhead template repeats the institutional banner at the top of every printed page (via a `<thead>` that Chrome repaints), applies the P→LP→VUS→LB→B colour scale to the primary find-

ing, and appends a technical appendix (QC metrics, methods summary, limitations and disclaimer) with a signature/stamp block for the authorising laboratory head.

8. Configuration

All thresholds and gene rules live in `familyforge/config/default.yaml`. Override any subset with `--config your.yaml`; your keys are merged on top of the defaults.

Key	Default	Meaning
`min_gq`	20	Minimum genotype quality; below → genotype not usable
`min_dp`	8	Minimum depth at the genotype
`min_alt_ab`	0.20	Minimum ALT allele-balance to accept a het/hom-alt
`max_ref_ab_for_hom`	0.10	Max REF fraction tolerated for a hom-alt
`af_dominant_max`	0.0001	Max popmax AF for a plausible **dominant** causal allele
`af_recessive_max`	0.01	Max popmax AF for a plausible **recessive** causal allele
`af_fields`	(list)	CSQ/INFO AF fields to take the max over (incl. African-ancestry)
`lof_consequences`	(list)	Consequences counted as loss-of-function
`missense_consequences`	(list)	Consequences counted as missense/protein-altering
`splice_region`	(list)	Splice-region consequences
`phylop_conserved`	2.0	PhyloP above this = conserved position
`revel_supportive`	0.5	REVEL $\geq$ this = supportive
`revel_strong`	0.7	REVEL $\geq$ this = strong (PP3-strong-ish)
`cadd_high`	25.0	CADD PHRED $\geq$ this = high deleteriousness
`mz_twin_groups`	`[]`	MZ-twin sample-ID groups (also settable via `--mz-twins`)
`mz_min_concordance`	0.97	Expected genome-wide MZ concordance
`stat3_dbd_range`	`[321,465]`	STAT3 DNA-binding-domain residue range
`stat3_sh2_range`	`[585,688]`	STAT3 SH2-domain residue range
`stat3_hotspot_residues`	(list)	Individual STAT3 hotspot residues

Key	Default	Meaning
`lof_mechanism_genes`	(list)	Genes whose LoF is the disease mechanism
`cnv_priority_genes`	`[DOCK8]`	Panel genes where deletions are especially important

The `af_fields` default deliberately includes African-ancestry fields (`gnomADe_AFR_AF`, `gnomADg_AFR_AF`, `gnomAD_AF_afr`) alongside the global ones, so frequency filtering is appropriate for African-ancestry families. The `stat3_*`, `lof_mechanism_genes`, and `cnv_priority_genes` keys are the **gene-specific rules**: they bias the tiering toward known disease mechanisms, and you extend them (or replace them for a different disease area) purely in configuration.

9. Interpretation logic in detail

This section documents exactly how a candidate becomes a tier and an evidence trail. It mirrors `acmg.tier_candidate` and the inheritance models; Figure 2 is the decision map.

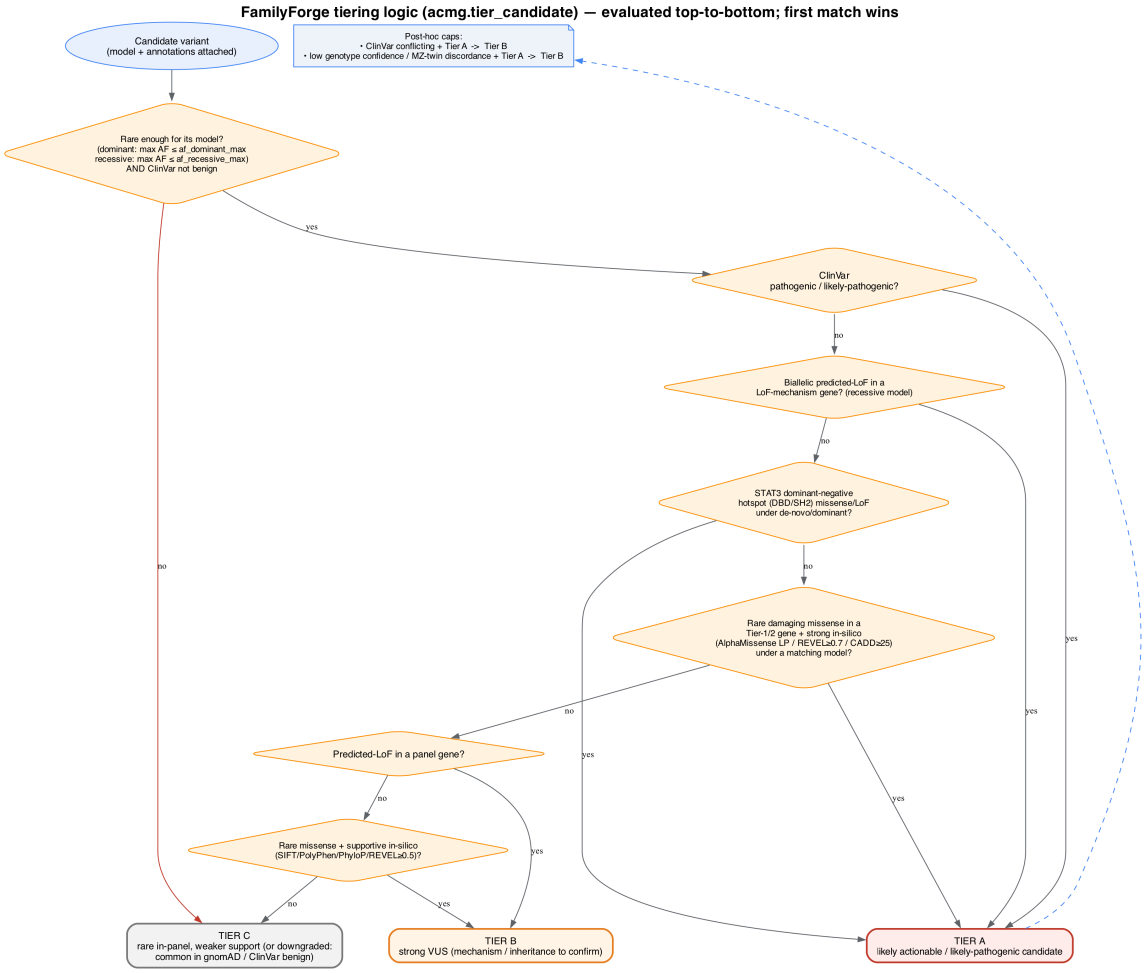


Figure 2: FamilyForge tiering decision flow (`acmg.tier_candidate`). Branches are evaluated top-to-bottom and the first match wins; two post-hoc caps demote a Tier A to Tier B under ClinVar-conflicting or low-confidence/MZ-discordant conditions.

Step 1 — establish the model. Every candidate arrives already carrying a model (`de_novo_or_dominant`, `ar_homozygous`, `ar_compound_het`, `x_linked_recessive`, or `cnv_deletion`) set by the inheritance/CNV step, together with a segregation string and a confidence (`high`/`moderate`/`low`).

Step 2 — gather evidence. For SNV/indel candidates, `tier_candidate` derives: ClinVar class (pathogenic / benign / conflicting / VUS / none), whether the consequence is LoF / missense / splice-region, whether the allele is **rare enough for its model** (using `af_dominant_max` for dominant models and `af_recessive_max` for recessive ones), whether the gene is a LoF-mechanism gene, whether it is a STAT3 hotspot, whether the position is conserved (PhyloP), and whether in-silico support is *supportive* or *strong* (SIFT deleterious, PolyPhen damaging, PhyloP conserved,

AlphaMissense likely-pathogenic, REVEL $\geq$ 0.5/0.7, CADD $\geq$ 25). It records a human-readable **evidence list** used verbatim in the reports.

Step 3 — decide the tier (first match wins):

1. **Not rare, or ClinVar benign → Tier C** (downgraded).
2. **ClinVar pathogenic/likely-pathogenic → Tier A** ("rare, segregates").
3. **Biallelic predicted-LoF in a LoF-mechanism gene, recessive model → Tier A.**
4. **STAT3 dominant-negative hotspot missense/LoF, de-novo/dominant → Tier A.**
5. **Rare damaging missense in a Tier-1/2 gene with strong in-silico consensus, under a matching-inheritance model → Tier A** (likely-pathogenic candidate; pending functional + parental confirmation).
6. **Predicted-LoF in a panel gene → Tier B** (mechanism/inheritance to confirm).
7. **Rare missense with supportive in-silico → Tier B.**
8. **Otherwise → Tier C.**

Step 4 — apply caps. A Tier A with *conflicting* ClinVar is demoted to Tier B; a Tier A with low genotype confidence (e.g. an MZ-twin discordance) is demoted to Tier B with the reason annotated.

How the reports read this. `report.classify()` translates (tier, ClinVar, AF) into the five-level P/LP/VUS/LB/B colour code; `rank_candidates` orders findings by tier, then confidence, then the gene's panel tier, so the single most-defensible finding is always `top`. The clinician and publication templates print the model, zygosity, segregation string, evidence trail, and tier reason; the letterhead engine renders the `top` finding as the primary result with recommendations.

10. Outputs

A `familyforge` run writes, into `--outdir`:

File	Audience	Contents
`clinician_report.html` / `.md`	Clinician / MDT	Concise: summary, family table, priority (Tier A/B) findings with evidence, clinical implications, QC, caveats
`publication_report.html` / `.md`	Manuscript / case report	Full: methods, tiered candidate tables (A/B/C), per-candidate evidence, provisional ACMG-code table, QC tables, panel table, limitations
`candidates.json`	Downstream / letterhead engine	Machine-readable: context summary, ranked rows, and QC — the input to `familyforge-clinical-report`

Running `familyforge-clinical-report` then adds `clinical_report.html` (+ `.md`), which you render to `clinical_report.pdf`.

Reading a candidate row. Each row carries: gene, model, tier, confidence, zygosity, segregation string, inheritance note, tier reason, the flags/evidence list, and (for SNV/indels) locus, rsID, filter, consequence, clinical-HGVS (MANE-prefixed), ClinVar, gnomAD (global + AFR), PhyloP, SIFT, PolyPhen, and CADD/REVEL/AlphaMissense where supplied. Compound-het rows also carry the partner allele; CNV rows carry the deletion locus and copy-number.

11. Report templates & the clinical letterhead

FamilyForge ships **three report styles**, all as Jinja2 templates under `familyforge/templates/`:

- `clinician.{html,md}.j2` and `publication.{html,md}.j2` — rendered by `report.py`. These are the concise and full research reports.
- `clinical_letterhead.{html,md}.j2` — rendered by `clinical_report.py`, the institutional letterhead document.

Honest note on disease framing. The **letterhead engine is fully disease-agnostic** — the disease name, panel name, methods wording, sample labels, and management text all come from the `report_config` YAML, so it produces a correct report for any disease. The **clinician and publication templates, by contrast, currently carry HIES-oriented framing text**: fixed titles ("…Suspected Hyper-IgE Syndrome"), a couple of gene-specific clinical- implication bullets (DOCK8 → HSCT; STAT3 → Job-syndrome management), and an illustrative methods paragraph naming a specific capture kit / VEP cache version. The *data* in these reports (candidates, tiers, evidence, QC, panel) is fully general and driven by your inputs; only that surrounding narrative is HIES-specific. If you run a non-HIES panel, prefer the letterhead report for clinical use, and treat the fixed narrative in the other two as a template to adapt. This is a documentation-honest limitation, not a defect in the analysis.

The letterhead template is designed for print: A4 geometry, a repeating banner header, the P→LP→VUS→LB→B colour scale, a variant-detail table, interpretation and recommendations, a signature/stamp block, and a second-page technical appendix (QC metrics, methods, limitations & disclaimer). Because logos are embedded as base64, the HTML is a single self-contained file that headless Chrome prints faithfully to PDF.

12. Worked example — the self-test

The bundled `tests/selftest.py` is the fastest way to see FamilyForge end-to-end. It writes a synthetic annotated VCF for a four-member family (`MOTHER`, `SIBLING`, and the affected MZ twins `TWIN_1`/`TWIN_2`), with an unsequenced `FATHER` in the PED, and plants four known-answer variants plus a benign decoy:

1. **STAT3** de-novo hotspot missense (p.Arg382Trp) — het in both affected, ref in mother/sibling → expected **Tier A**, de-novo/dominant.
2. **DOCK8** homozygous stop-gained — hom in both affected, het in mother/sibling → expected **Tier A**, AR-homozygous.
3. **PGM3** compound-het (a maternal stop-gained + an inferred-paternal frameshift) → expected **Tier A**, AR-compound-het.
4. A common, benign **STAT3** missense → expected to be filtered out (downgraded).

```
python tests/selftest.py
```

Expected result:

```
=== SELFTEST RESULT ===
Tier A genes: {'STAT3', 'DOCK8', 'PGM3'}
summary: {'n_candidates': ..., 'tierA': 3, 'tierB': ..., 'tierC': ...}
...
SELFTEST PASS
```

The test also confirms the four report files are written and non-empty, and that the MZ pair `TWIN_1~TWIN_2` shows high genotype concordance. Inspect the generated `out/` directory to see real clinician, publication, and JSON outputs.

13. Troubleshooting

Symptom	Cause / fix
`WARN: bcftools not on PATH - skipping normalisation`	Install `bcftools`, or pass `--skip-norm` if the VCF is already biallelic-split. Without it, multiallelic sites are ignored.
`loaded 0 panel variants`	The picked-consequence gene never matched the panel. Check that `SYMBOL` in the VEP `CSQ` uses the same gene symbols as the panel, and that the VCF is actually annotated.
A sequenced member shows as *not sequenced*	Its PED `iid` (column 2) does not exactly match a VCF sample name. Names are case-sensitive.
No Tier A/B despite an obvious candidate	Check genotype quality (a low-`GQ`/`DP`/`AB` call is gated out), the panel's `inheritance` (an incompatible model will not fire), and the AF thresholds (too-common alleles are downgraded).
Expected de-novo call is "unconfirmed"	This is by design when the father is not sequenced — add paternal DNA to confirm.
MZ pair flagged discordant / Tier A capped to B	A genotype differs across the twins at that site — usually a technical dropout; review depth/AB, and confirm the twins really are the declared pair via the QC concordance table.
CNV rows ignored	The row was not recognised as a deletion (call text / copy-number / log2 thresholds), or its gene did not overlap the panel. Check column names against §5.5.
Letterhead engine errors on a missing logo	`logo_banner`/`logo_lab` paths in the lab-config must point to readable image files (they are base64-embedded).

14. Reproducibility & provenance

Reproducibility rests on three things: the **inputs** (VCF, PED, panel, config), the **configuration** (all thresholds and rules are in `config/default.yaml` and echoed into the publication report's methods), and the **upstream provenance** (`--provenance` threads CallForge's tool/DB versions and git commit into the report). The `candidates.json` is a complete, machine-readable record of every ranked candidate and the QC, so a run can be re-examined without re-running the analysis. Pin the package version (`familyforge.__version__`, printed in every report) and the config file alongside your results.

15. Limitations

- **Candidates, not diagnoses.** Every Tier A/B finding is pending orthogonal confirmation (Sanger / MLPA / aCGH), functional validation, and formal ACMG classification by an accredited laboratory.
- **Missing parents.** With an unsequenced parent, de-novo status and compound-het phase are *inferred*, not confirmed.
- **Annotation-bound.** Interpretation quality is bounded by the upstream annotation: population-AF completeness (ancestry-specific fields especially), and in-silico coverage. When genome-wide dbNSFP is unavailable, supply finalist scores via `--extra-scores`.
- **Panel-restricted.** The primary pass keeps only variants in panel genes; variants outside the panel are not evaluated.
- **Pragmatic tiering.** The A/B/C rubric is a triage signal, not the full ACMG framework; it does not enumerate every ACMG criterion or compute a formal combined classification.
- **Report framing.** The clinician/publication templates carry HIES-oriented narrative (§11); the letterhead engine is disease-agnostic.
- **CNV depth.** CNV cross-check surfaces deletions overlapping panel genes from an external caller's output; it does not itself call CNVs and requires orthogonal confirmation.

16. Glossary

- **ACMG** — American College of Medical Genetics variant-classification framework; FamilyForge implements a pragmatic *subset* for triage.
- **AF / popmax** — population allele frequency; the maximum across the configured `af_fields`.
- **Compound heterozygous (in trans)** — two different heterozygous variants in the same gene on opposite parental haplotypes.
- **De novo** — a variant present in the child but neither parent; here often *inferred* when a parent is unsequenced.
- **Hemizygous** — a single-copy genotype (males on chrX/chrY).
- **HGVS** — standard nomenclature for cDNA (`c.`) and protein (`p.`) changes; reports prefer the MANE Select RefSeq transcript.
- **LoF** — loss-of-function (stop-gained, frameshift, canonical splice, etc.).
- **MANE Select** — the jointly agreed representative transcript per gene (NCBI/EMBL-EBI).
- **MZ twins** — monozygotic (identical) twins; treated as genetic replicates.
- **Segregation** — how a variant's genotypes track the affected/unaffected pattern in the family.
- **VUS** — variant of uncertain significance.

17. References & citation

Cite FamilyForge using the repository `CITATION.cff`:

> Adu, B. *FamilyForge: pedigree-aware Mendelian and ACMG-style interpretation of a family callset.* Genomics & Bioinformatics Core, Noguchi Memorial Institute for Medical Research, University of Ghana, Legon, Ghana. Version 0.1.0, 2026.

The forge family. CaptureForge (design) → CallForge (analysis) → FamilyForge (interpretation). FamilyForge consumes CallForge's annotated joint VCF, `cnv_calls.tsv`, and `provenance.json`.

Underlying resources referenced by the interpretation logic and reports: Ensembl VEP (consequence, HGVS, MANE, SIFT, PolyPhen); gnomAD (population allele frequencies); ClinVar (clinical significance); PhyloP (conservation); and the finalist in-silico predictors CADD, REVEL, and AlphaMissense. The ACMG/AMP variant-interpretation guidelines (Richards et al., 2015) inform the pragmatic tiering rubric.